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A Preliminary Investigation of Cognitive Intolerance and Neuroimaging among Adolescents Returning to School after Concussion

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Abstract

Primary Objective: To introduce the concept of cognitive intolerance. A test is proposed to measure this concept and pilot data are presented to support this measure and future research to develop this concept into a construct.

Research Design: Three group comparison to protect larger study blinding.

Methods and Procedures: Two groups of student athletes ($n=13$, $n=13$) between 13 and 17 (mean 15.1 ± 1.1 years; 58% male) who sustained a sport related concussion within 10 days and one group ($n=13$) of age matched healthy controls were recruited for a comparison of correlations between self and observer ratings of cognitive difficulties and DTI fractional anisotropy (FA) using Tract-Based Spatial Statistics (TBSS) analysis at two time points.

Main Outcomes and Results: Significant negative only associations (higher cognitive difficulty and lower FA) with DTI FA were found in white matter tracts. These included the anterior corpus callosum, frontal parietal longitudinal fasciculi, and cortical subcortical pathways at only the second time point. Several working memory networks would likely involve connections using the above identified white matter tracts.

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Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conclusions: Cognitive intolerance can be defined as symptom exacerbation from prolonged cognitive activity. Cognitive intolerance could be measured by the n-back working memory task and time to symptom exacerbation.

Keywords

Concussion; cognitive intolerance; neuroimaging; neuropsychology; return to learn

Adolescents returning to school after a sport-related concussion (SRC) often experience symptom exacerbation with prolonged cognitive activity. This difficulty could be referred to as “cognitive intolerance.” Thus, the concept of cognitive intolerance can be defined as symptom exacerbation from prolonged cognitive activity. This kind of difficulty may occur even after students have successfully returned to exercise and sports. The concept of cognitive intolerance can be measured by time to symptom exacerbation during a cognitive task. Neuroimaging literature and pilot data investigating aspects of brain dysfunction that are associated with cognitive difficulties during concussion recovery provides preliminary support for a measure of this concept and for developing this concept into a more formal construct. Establishing the construct of cognitive intolerance may help to develop interventions to improve return to school and other activities after concussion.

Corwin et al (2014) (1) in a sample of 247 patients aged 5 to 18 years found that 73% needed accommodations at school and 61% experienced a decline in grades after concussion. The median time to return to school full time without accommodations was 35 days (IQR 11–105). Seventy-three percent took longer than 4 weeks to recover and the median time to becoming symptom free was 64 days (IQR 18–119). Another study (2) found that severity of concussion symptoms was associated with school-related problems and with worse academic effects experienced by students aged 5 to 18.

Activity level may be associated with cognitive difficulty. Carson et al (2014) reported that 45% of students returned to school too soon and experienced an exacerbation of symptoms (3). A recent prospective study of 355 student athletes found that those in the highest quartile of cognitive activity experienced a longer duration of symptoms, and the authors recommended limiting extensive cognitive activity. They noted reduced CBF and the possible mismatch between cell needs and perfusion as a possible explanation for this symptom exacerbation (4). Other studies have found associations between higher cognitive activity levels and longer recovery times (5, 6). Makki, et al (2016) found an association between the number of hours a student spent in school each day after a concussion and daily symptom levels (7).

Exercise Intolerance:

The Buffalo Concussion Treadmill Test (BCTT) was developed to systematically determine exercise intolerance after concussion (8, 9). Using the BCTT, the threshold of exercise tolerance is determined at the heart rate when an individual experiences exacerbation of symptoms. An exercise program at a heart rate below this symptom threshold has led to improved symptoms and eventual recovery of exercise tolerance in several studies, including a randomized clinical trial (10–12).

Cognitive intolerance:

An exacerbation of symptoms can also occur with prolonged cognitive activity (13). The exacerbation of symptoms from cognitive activity appears to be related to length of time and intensity of cognitive activity (14, 15) rather than to heart rate. A key aspect of developing the concept of cognitive intolerance into a more formal construct involves distinguishing exacerbation of symptoms due to prolonged cognitive activity from performance scores obtained on brief computerized cognitive testing. Another aspect involves eventually distinguishing cognitive intolerance from traditional cognitive impairment based on relatively brief performance on tests of neuropsychological domains such as attention and memory, and establishing a measure of cognitive intolerance.

Although performance on brief computerized cognitive tests is widely used to measure recovery after concussion, there is considerable debate over their usefulness in determining recovery. This may be due in part to differences between cognitive performance scores on this type of relatively brief task and symptom exacerbation over an extended period of time from the type of inefficient working memory processing that has been found in fMRI studies during the n-back task (16–18), and that can persist for at least two months (19).

Nelson et al. 2016 (20) studied the usefulness of three widely used brief computerized tests. The sensitivity of reliable change indices to distinguish between patients and non-injured controls was best at 24 hours after injury, but declined to near the false positive rate after that. Over repeat testing at 8, 15, and 45 days, sensitivities were best within 1 day of becoming asymptomatic but were similar to false positive rates for athletes who became asymptomatic several days earlier.

Neuroimaging and Neurophysiological Mechanisms:

Developing a measure of the concept of cognitive intolerance and then developing this concept into a more formal construct involves the neurophysiological mechanism that underlies cognitive intolerance. It is possible that initial cognitive intolerance and exercise intolerance share a similar neurophysiological mechanism during the typical recovery period (up to about 4 weeks after injury). Diffuse and/or focal neuronal dysfunction due to stretching injury to neuronal axons may contribute to Autonomic Nervous System (ANS) dysfunction and altered cerebral blood flow (CBF) regulation during both exercise and cognitive activity (21). Tan et al (2014) (22) suggest that cerebrovascular function is determined by vasoreactivity, autoregulation, and neurovascular coupling. Altered CBF regulation due to changes in vasoreactivity and autoregulation from ANS dysfunction may produce neurovascular uncoupling, leading to cognitive difficulties soon after injury (22). This mechanism may explain exercise intolerance and could explain the early recovery of tolerance for cognitive activities within the first 4 weeks. Continued symptom exacerbation with prolonged cognitive activity after about 4 weeks may involve a relatively larger contribution from diffuse and/or focal neuronal dysfunction due to stretching injury to neuronal axons compared to ANS dysfunction (23, 24) (25, 26).

A number of studies have shown findings on DTI imaging during early and later periods of recovery from SRC and mild TBI (mTBI). Henry et al (2011) (27) found that fractional anisotropy (FA) on DTI imaging was increased and mean diffusivity (MD) was decreased in the corticospinal tracts and the corpus callosum at 1–6 days and again at 6 months after SRC. Cubon et al (2011) (28) used DTI tract-based spatial statistics (TBSS) at 1 month after injury and concluded that TBSS analysis is sensitive enough to detect structural changes after SRC. Lipton et al (2012) (29) used a voxel-wise approach at 2 weeks, 3 months, and 6 months after injury. The spatial pattern of high and low FA varied among patients. The authors attributed individual differences in FA to anatomy, vulnerability to injury, and the mechanism of injury. They noted that areas of high FA may represent a compensatory mechanism or plasticity in response to injury. Results from a meta-analysis by Eierud et al (2014) (30) supported the hypothesis that acute mTBI (< 2 weeks) is associated with elevated FA values while chronic mTBI complaints (> 2 weeks) are correlated with depressed FA.

Studies have also found associations between DTI findings and aspects of cognition, including working memory and processing speed. Both of these aspects of cognition may contribute to cognitive intolerance. Xiong et al (2014) (31) found that elevated MD in the uncinate fasciculus was significantly correlated with the working memory index and the processing speed index of the Wechsler Adult Intelligence Scale (WAIS-IV) at 1 month after injury. Lipton et al (2009) (32) used whole brain voxel-wise analysis and found multiple clusters of lower FA in the dorsolateral prefrontal cortex with higher MD in some of these clusters at 1 month after injury. Lower FA was correlated with worse executive function scores. Executive functioning includes working memory and involves the frontal cortex. Niogi et al (2008) (33) found that reaction time was correlated with the number of white matter structures with reduced FA between 1 and 65 months after injury. The Eierud et al (2014) (30) meta-analysis found that correlations between FA and neuropsychology test performance changed from negative before two weeks to positive after two weeks post injury.

These studies provide support for the hypothesis that diffuse and/or focal neuronal dysfunction that varies across individuals and may vary across time plays a role in persistent cognitive difficulties. Difficulties with working memory and other aspects of executive function, as well as processing speed, may contribute to persistent cognitive difficulties, which may have a contribution from axonal dysfunction as measured by DTI.

The purpose of this study was to introduce the concept of cognitive intolerance and to present pilot data in support of a measure of this concept and future research to develop this concept into a more formal construct. Development of the cognitive intolerance concept would allow for improved diagnosis and return to learn decisions. A cognitive intolerance test would aid in measuring the outcomes of interventions for cognitive difficulties, such as cognitive pacing, cognitive rehabilitation, and cognitive behavioral therapy, and aid in developing new interventions to help with returning to school and other activities.

METHODS

Participants:

Participants were recruited from patients in a concussion management clinic as part of a larger study. The sample included males (n=15) and females (n=11) aged 13 to 17 years (Mean 15.1±1.1 years) who sustained a concussion during sport (competitive or recreational) within the previous 10 days. Exclusion criteria included: non-sport mechanism of injury; GCS score <13 at any time post injury or lesion on CT/MRI (via review of medical records or if identified by study imaging); focal neurologic deficit on physical examination; major depression or another psychiatric or neurologic disorder; or a history of one or more other concussions within the year prior to the first visit. A separate group (n=13) was also recruited of age matched healthy controls who had not sustained a concussion. The study was approved by the University Institutional Review Board (IRB).

Measures:

Scores for intolerance of cognitive activity were obtained based on 0 to 6 patient ratings of cognitive symptoms from the Post Concussion Symptom Scale (PCSS) used in our clinic (34) and from a chart extraction from physician notes using a pre-determined algorithm. Three scores for intolerance of cognitive activity (C-1, C-2, and C-3) were created for use in the TBSS analyses from the patient's third clinic visit (mean 22.4±4.2 days post injury, range 18 to 30 days). The cognitive scores were derived as follows: (1) C-1 is the 0 to 6 patient rating of concentration difficulties from the PCSS, since patients often describe difficulties with working memory processing as, "concentration difficulties;" (2) C-2 is the sum of the 0 to 6 ratings of 7 symptoms relevant to cognitive intolerance (concentration, memory, feeling slowed down, answering slowly, fatigue, foggy, and confused); (3) C-3 is a dichotomous rating completed by a neuropsychologist that rates intolerance for cognitive activity based on a predetermined decision rule from physician chart notes and the seven symptoms noted above. A patient was considered intolerant if one or more indications (school difficulties, school accommodations, cognitive intolerance, and/or cognitive pacing) was mentioned in the chart notes and the sum of the seven cognitive symptoms score was more than 2. Otherwise, they were not considered intolerant. Although these variables were created with some overlapping inputs and are to some degree collinear, they represent subjective and observer ratings of cognitive difficulties.

Procedures:

Potential participants were examined by a physician experienced in concussion evaluation. Concussion diagnosis by the physician was made using current criteria, including a standardized neurological examination and the SCAT-5 instrument if available, which combines symptoms, cognitive performance and balance (35). If a participant satisfied the inclusion/exclusion criteria, they were consented for participation. The same procedure was followed for control subjects.

We carried out a voxel-wise statistical analysis of the diffusion fractional anisotropy (FA) data using TBSS (36). First, FA images were created by fitting a tensor model to the raw diffusion data using FDT, and then brain-extracted using BET (37). All subjects' FA data

were then aligned into a common space using the nonlinear registration tool FNIRT (38). Next, the mean FA image was created and thinned to create a mean FA skeleton, which represents the centers of all tracts common to the participant group. Each participant's aligned FA data was then projected onto this skeleton and the resulting 4-dimensional data was assessed using non-parametric permutation testing. Individual cognitive scores (C-1, C-2, and C-3) were then regressed against FA values while controlling for age and sex.

In order to protect blinding for the larger study, we divided the patients and controls ($n = 39$) into three equal groups; Group A is the first 13 patients, Group B is the 13 controls, and Group C is the other 13 patients. The patients (Groups A and C) have their actual three cognitive intolerance scores. The controls in Group B have random scores that were selected from the patients' scores.

Figures showing the results of the TBSS analysis were created by regressing the three cognitive scores (C-1, C-2, and C-3) for each of the three groups (A and C, A and B, B and C) for both negative and positive associations (high C score and low FA value, Low C score and low FA value, respectively) at the follow-up imaging time point, which occurred as soon as possible after the third clinic visit. This resulted in a total of 18 figures at the follow-up imaging time point ($3 \text{ scores} \times 3 \text{ groups} = 9 \times \text{positive and negative associations} = 18 \text{ figures}$). Five of these figures (Figures 1 through 5) are shown and the remaining 13 figures are reported.

For the initial imaging time point after the first clinic visit about a week after injury only one of the scores (C-1) was regressed for all three groups in the negative and positive direction resulting in 6 figures ($3 \text{ groups} \times \text{positive and negative associations} = 6 \text{ figures}$). One of these figures (Figure 6) is shown and five are reported.

RESULTS

The average recovery time for patients ($n = 26$) was 25.0 ± 16.8 days (range 8 to 75). At an average of 22.4 ± 4.2 days post injury (range 18 to 30), 10 (38%) showed at least some degree of intolerance of cognitive activity based on the symptom scores and physician assessment.

Figures 1, 2, and 3 show results of the TBSS analysis investigating the relationship between the C-1, C-2, and C-3 scores, respectively, for the 26 patients (Groups A and C) and the voxel-wise FA values at the follow-up imaging time point. The group FA skeleton is shown in green and regions significantly ($p < 0.05$) related to the scores are shown in red/yellow. Significance values are corrected for family-wise error rate due to multiple comparisons. Significant associations in these three figures are in the negative direction (higher cognitive symptom scores and lower FA values).

Figure 1 shows the association between C-1 scores (patient reported concentration difficulties) and FA values along white matter tracts. Figure 2 shows this association for C-2 scores (the sum of seven symptoms relevant to cognitive difficulties). Figure 3 shows the association between C-3 scores (dichotomous rating of intolerance of cognitive activity or not based on physician chart notes and patient reported symptoms) and FA values.

Each of these three figures show significant negative associations at the follow-up imaging time point in numerous areas of the FA skeleton, including the anterior corpus callosum (genu) connecting right and left frontal lobe regions, longitudinal fasciculi connecting parietal lobe (posterior) and frontal lobe (anterior) regions, and ascending/descending tracts connecting cortical and subcortical regions. In addition, it is important to note that there were no significant positive associations at the final imaging time point found on the nine figures for any of the three C scores (C-1, C-2, or C-3) for any of the three groups (AC, AB, BC).

Figure 4 shows the association between C-1 scores (patient reported concentration difficulties) and FA values for a mixture of 13 patients and the 13 controls (Groups A and B) at the follow-up imaging time point. Figure 5 shows C-3 scores (yes/no cognitive intolerance rating) for a mixture of the other 13 patients and the 13 controls (Groups B and C) at the follow-up time point. Both of these figures show the green white matter FA skeleton only and no significant negative associations. In addition, the remaining 4 figures (not shown) for the other cognitive scores for the A and B and the B and C Groups show only the green white matter FA skeleton and no significant negative associations at the follow-up imaging time point.

Figure 6 shows the association between C-2 scores (sum of 7 cognitive symptoms) and FA values at the initial imaging time point for the 26 patients (Groups A and C). As seen in Figure 6, in contrast to the final imaging time point, there are no significant negative associations (high C-2 score and low FA value) at this initial imaging time point. There were also no other significant negative or positive associations on the remaining five figures found at the initial imaging time point for the C-2 score and FA values for any of the three groups.

The DTI mean FA values for the 26 patients (Groups A and C) showed a slight decrease from the initial to the final imaging time points for white matter, grey matter, whole brain, insula, brainstem, and parietal, temporal, and occipital lobes.

DISCUSSION

Using Tract Based Spatial Statistics, we found significant associations in a number of white matter tracts between several measures of cognitive difficulty and DTI fractional anisotropy measures during recovery from concussion. These white matter tracts included the genu of the anterior corpus callosum connecting hemispheres of the frontal lobes and other tracts connecting frontal lobe regions to parietal lobe regions and subcortical structures. Several working memory networks would likely involve connections using these tracts, including the frontal executive network (FEN) involving right and left hemispheres of the frontal lobes, the default mode network (DMN) involving the posterior cingulate/precuneus region of the parietal lobes, and connections between these two networks involving left and right frontal parietal networks (FPN), as found on standard space anatomical maps.(39)

These associations were found only for the third clinic visit and second imaging time point while associations with one of the scores were not found at an initial clinic visit and first imaging time point. Only negative associations (higher cognitive difficulty and lower FA

values) were significant. When these analyses were repeated with a group composed of half patients and half healthy controls, the associations did not rise to the level of statistical significance. In addition, the overall mean values for FA in the concussed group decreased slightly from the initial to the final time point for overall white matter and for most other brain regions, consistent with the findings of Eierud et al (2014).(30) This supports the hypothesis that acute mTBI (< 2 weeks) is associated with elevated FA values while chronic mTBI complaints (> 2 weeks) are correlated with depressed FA.

These pilot results provide some initial support for research into developing the construct of cognitive intolerance. As noted above, cognitive intolerance can be defined as exacerbation of symptoms with prolonged cognitive activity. This prolonged activity may especially involve working memory processing or “concentration.” An important step in developing the construct of cognitive intolerance will be to develop a measure of cognitive intolerance that is associated with the brain dysfunction that contributes to symptom exacerbation with prolonged cognitive activity and that is associated with difficulties in returning to school and other activities.

As noted above, performance scores on brief computerized tests have been shown to be useful only within 24 hours of injury and within a day of becoming asymptomatic. An alternative to currently used brief computerized tests may be to assess a patient’s tolerance for cognitive activities using a measure such as the Children’s Exertional Effects Rating Scale (40) (CHEERS), developed by Gerald Gioia, Ph.D., and administered before and after a cognitive task. The CHEERS measures increase in symptoms (headache, fatigue, concentrating problems, irritability) after a cognitive task and it has good reliability and validity for difficulty in school (41).

In addition to the overall effects of slowed processing speed after concussion, inefficient working memory processing may contribute to symptom exacerbation during prolonged cognitive activity. This may be reported by patients as “concentration difficulties.” Time to symptom exacerbation during the n-back task may provide an alternative to widely used computerized tests of brief cognitive performance. This measure of cognitive intolerance may also provide an alternative to assessment of cognitive impairment that measures degree of departure from norm-referenced performance on tests of traditional neuropsychological domains such as attention, memory, and executive function.

The n-back task has been studied extensively as a test of working memory and with respect to its neurophysiological basis. Most of this research has studied the blood oxygenation level dependent (BOLD) fMRI response during the n-back task. Kearney-Ramos et al (2014) (42) supported the construct validity of the n-back task as a test of working memory. Using fMRI, they identified 17 neural networks that demonstrated n-back engagement. Eight networks, including bilateral dorsolateral prefrontal cortex and bilateral frontal parietal networks, were recruited and indicated working memory involvement. Their results supported the use of the n-back task in probing network level neural correlates of working memory processing. Abnormal working memory may be an important contributor to the inefficient neural processing that leads to symptom exacerbation during prolonged cognitive tasks, i.e., cognitive intolerance.

The n-back task also allows for a gradual increase in difficulty that can be extended over a period of time to allow for assessment of symptom exacerbation. McAllister et al (2001) (43) used fMRI to study increasing working memory load on the auditory n-back task. While controls showed increased activation in regions of working memory circuitry with each increase in processing load, concussed patients disproportionately increased activation during moderate load (2-back) with very little increase with the highest load (3-back). Their results suggested injury-related changes in the ability to modulate working memory processing resources. These results also suggest the potential usefulness of the n-back task when testing for tolerance of cognitive activity since it distinguished between concussed patients and controls with increased difficulty (and time).

Dettwiler et al (2014) (19) used the n-back task with fMRI to study longitudinal recovery after SRC. Athletes compared to controls showed persistent increased activation at 2 days, 2 weeks, and 2 months after injury with no performance differences. Most athletes were symptom free and had normal brief computerized testing and traditional neuropsychology testing by 2 to 3 weeks after injury; however, at 2 months there remained increased activation in the dorsolateral prefrontal cortex. The authors proposed three hypotheses to explain these activation changes: additional recruitment of neural resources to support task performance (neural compensation); brain reorganization from injury; and increased cognitive and attentional control similar to controls.

Other studies have used working memory tasks and fMRI to study brain activity during persistent post-concussive symptoms (PPCS). Chen et al (2008) found no initial differences between three groups of subjects (recovered, did not recover, and normal control) in behavioral performance using a working memory task. The PPCS groups had decreased activation in a specific region - the left dorsolateral prefrontal cortex (Brodmann area 9/46), a region where normal controls had increased activation while performing the working memory task. The PPCS groups also had activations in other areas not used by normal controls in performing the task. The authors interpreted this as compensatory recruitment of other brain regions to carry out the task (16).

A Cognitive Tolerance Test:

Taken together, these studies provide support for development of the n-back task as a test of inefficient working memory processing, and for inefficient working memory processing as a possible mechanism contributing to intolerance of cognitive activity, including symptom exacerbation after extended cognitive activity involving working memory. A cognitive intolerance test (CIT) would need considerable further development, including investigating the safety and any detrimental effects of symptom exacerbation, for use with patients recovering from concussion. A CIT could include time to symptom exacerbation using the n-back task. The n-back task could include, for example, a 5 minute duration for each of the 1-back, 2-back, and 3-back conditions. The CIT could be stopped at a significant exacerbation of symptoms (for example, this could be defined as ≥ 3 point change from the pre-task resting overall symptom score on a 1–10 point visual analog scale, VAS), at 4 non-responses or incorrect responses, or at patient request. Time to stopping point could provide a score on the CIT. Tolerance during the CIT measured by the difference between before and

after CHEERS scores may provide another score for a measure of tolerance of cognitive activity that would generalize to difficulties with prolonged concentration in school.

Mayer et al (2015) (44) noted that studies that combine information across various neuroimaging modalities can capture the multifaceted pathology of concussion, and they suggested that multimodal imaging studies with a longitudinal design could address recovery trajectories. Neuroimaging measures other than DTI and task fMRI may contribute to our understanding of intolerance of cognitive activity. These measures may include magnetic resonance spectroscopy (prefrontal hypometabolism) and resting state fMRI (hyperactivation of non-working memory networks and hyperconnectivity with the default mode network). Dean et al (2013) (45) used DTI, Magnetic Resonance Spectroscopy (MRS), and Functional MRI (fMRI) imaging with working memory (concentration) and attention/speed of processing tasks among 8 participants with PPCS, 8 non-PPCS participants, and 8 control participants more than 12 months post injury. Participants with PPCS showed reduced structural integrity in the corpus callosum and anterior white matter on DTI and reduced creatine concentration (relevant to energy metabolism) in the right dorsolateral prefrontal cortex on MRS. These findings correlated with increased functional fMRI activation in the anterior cingulate along with reduced activation in the default mode network and working memory areas (left prefrontal) as cognitive load was increased. The authors interpreted these results to suggest that top-down attentional regulation and deactivation of task-irrelevant areas may be compensating for the reduction in working memory capacity and variation in white matter transmission caused by structural and metabolic changes after injury. This compensation may in turn be contributing to PPCS. Hu, et al (2013) (46) found that high glutamate concentration in the posterior cingulate cortex / precuneus, a key component of the default mode network, was associated with subsequent reduced deactivation observed during the n-back task in the same region. Other studies have also found support for the use of MRS with concussion (47–50). In addition to the DTI and MRS studies noted above, other studies have also noted concussion-related impairments that persist well beyond 6 months when assessed by more complex cognitive tasks (51, 52). Most of these studies have included fMRI imaging during the n-back task, although Ozen et al 2013 (53) found inefficient recruitment of processing resources more than 6 months after concussion using the n-back task with EEG event-related potentials (P300 amplitude).

Mayer et al (2015) (44) also noted that resting state fMRI is well poised to be used to investigate connectivity after concussion. Studies have used resting state fMRI to determine how concussion may affect connectivity between different regions, including connectivity in the default mode network (DMN). Mayer et al, 2011 (54) used resting state fMRI to examine functional connectivity within 3 weeks and again at 3–5 months post injury. They compared spontaneous fluctuations in the DMN and the frontal parietal task-related networks between patients and controls. They found decreased functional connectivity within the DMN and increased connectivity between the DMN and the lateral prefrontal cortex within 3 weeks after concussion even though traditional neuropsychology assessment did not differentiate patients from controls. Functional connectivity classified 84% of cases and predicted cognitive complaints, although there were no significant changes in connectivity among a subset of patients who returned at 3–5 months. The authors noted that increased functional connectivity in regions associated with top-down attentional processing

along with decreases in the DMN may help to explain distractibility and cognitive fatigue. Zhou et al 2012 (55) also noted alterations in functional connectivity between 3 and 53 days (mean 22 days) after concussion. They found reduced connectivity in the posterior hubs of the DMN and increased connectivity within the ventromedial prefrontal cortex, with associations with cognitive inflexibility and symptoms. Another study (56) found that patients with persistent symptoms showed less deactivation of the DMN, especially during difficult tasks, and that functional connectivity between the DMN and the frontal executive network was higher.

Tolerance of cognitive activity early on may be improved through graded subsymptom threshold aerobic exercise treatment by normalizing CBF dysfunction. As noted above, it is possible that the mechanism of intolerance of cognitive activity during the early recovery period involves more ANS and CBF dysfunction compared to axonal dysfunction (21, 27, 33). Tan et al 2014 (22) note that neurovascular uncoupling (failure to increase CBF in response to increased neural activity) may explain cognitive difficulties during early subacute recovery up to about 4 weeks after injury. While this mechanism could also contribute to more persistent cognitive difficulties, an alternative hypothesis invokes diffuse axonal dysfunction and in some individuals more focal axonal dysfunction (23, 24, 26) due to stretching injury to neuronal axons. Delayed recovery of ANS dysfunction may also contribute to persistent cognitive difficulties (25).

This initial pilot study has limitations. The measures of cognitive difficulty were based on patient report of symptoms and chart extraction, and are subject to the many limitations of these types of data. Cognitive difficulties were assessed early on, an average of three weeks after injury, and continued recovery may change the results. The sample size is relatively small. Due to the necessity of preserving blinding for the larger study, the comparison groups were composed of a mixture of patients and healthy controls. Nonetheless, these pilot results lend some support to the proposed measure and to continued research into developing the construct of cognitive intolerance. The data reveal negative associations between cognitive difficulties and DTI FA values along white matter tracts that may be involved in working memory networks.

Future Research:

Understanding more about the neurophysiological mechanisms underlying tolerance of cognitive activity after concussion could help with developing a measure for this construct and establishing the reliability and validity of this measure, including neuroimaging correlates and correlates with school functioning. This would be an important step in understanding the neurophysiological basis for persistent problems with concentration, symptom exacerbation, and fatigue in school after concussion and in developing treatment approaches based on understanding the neurophysiological mechanisms. Future research could also include measuring the response to interventions for cognitive intolerance, such as cognitive pacing (limiting concentration to periods of time less than a cognitive symptom threshold) (14, 15, 57), cognitive rehabilitation, cognitive behavioral therapy, and new interventions to address the effects of slowed and inefficient neural processing.

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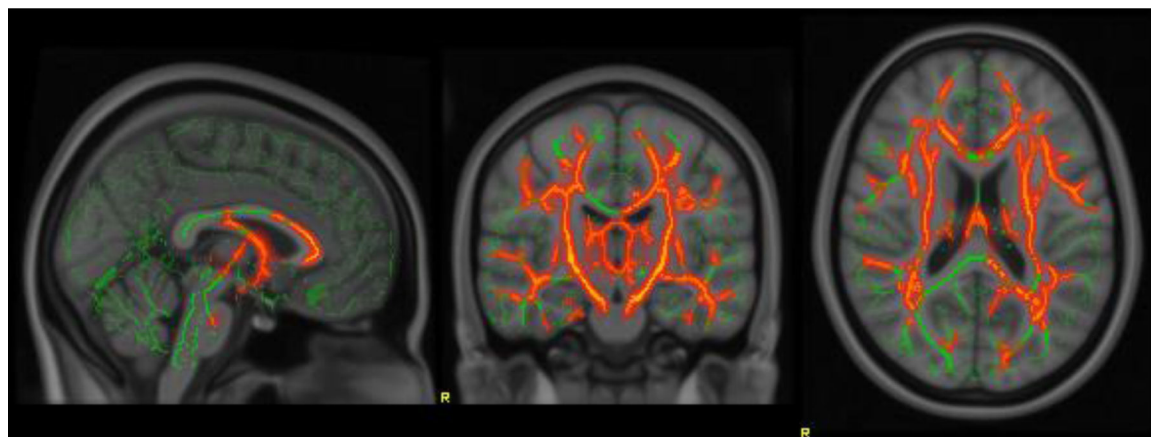


Figure 1.
Associations between greater patient-reported concentration difficulties and decreased FA values

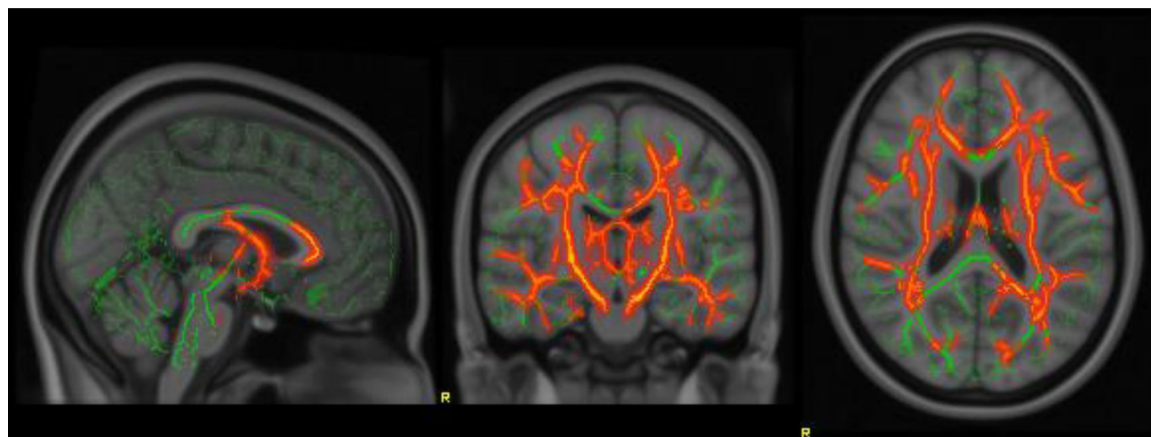


Figure 2.
Association between greater symptoms relevant to cognitive difficulties (n=7) and decreased FA values

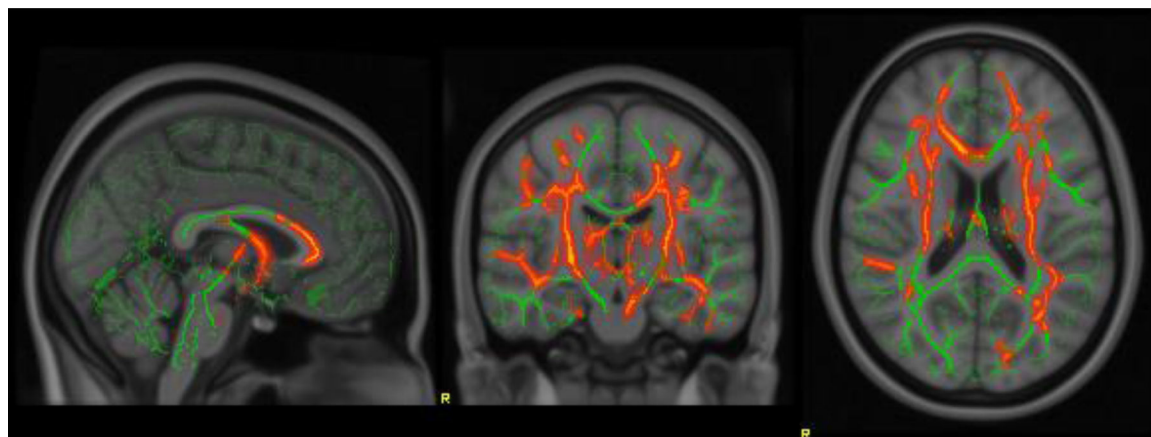


Figure 3.
Associations between dichotomous rating of presence of intolerance of cognitive activity and decreased FA values

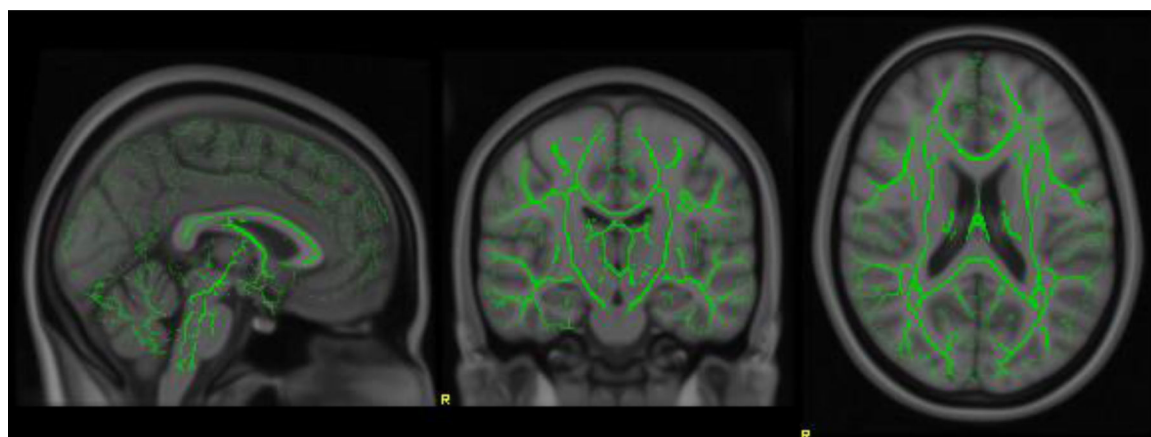


Figure 4. Non-significant associations between greater patient-reported concentration difficulties and decreased FA values for a mixture of 13 patients and 13 controls (Groups A and B) at the follow-up imaging time point

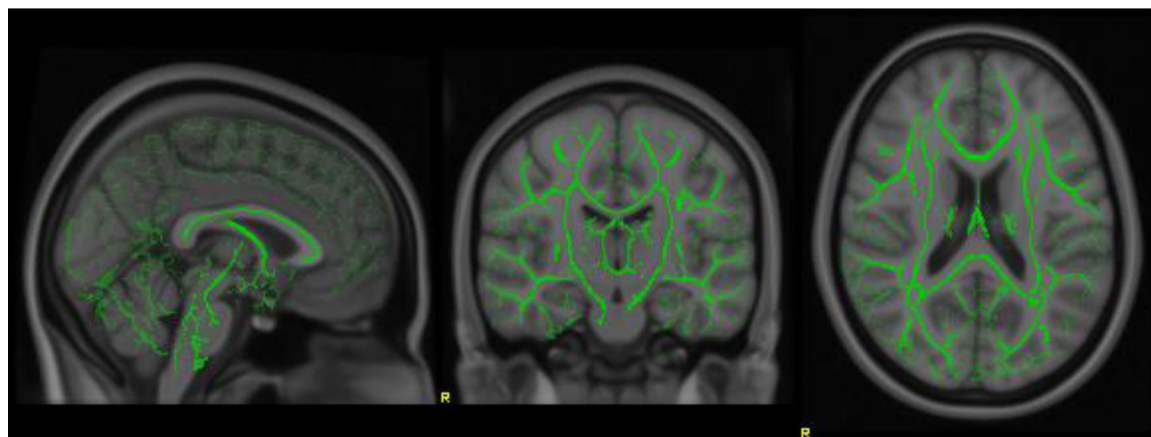


Figure 5. Non-significant associations between rating of presence of intolerance of cognitive activity and decreased FA values for a mixture of 13 patients and 13 controls (Groups B and C) at the follow-up imaging time point.

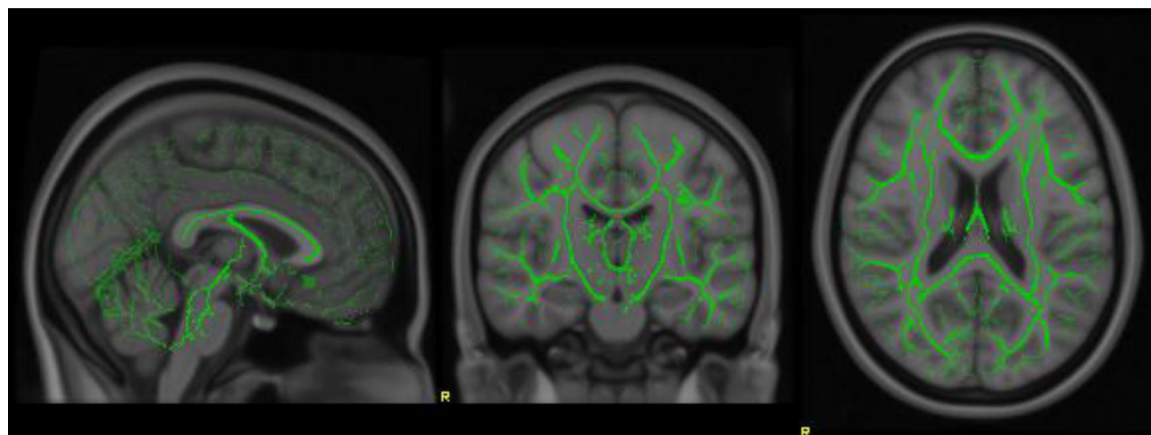


Figure 6. Non-significant associations between greater symptoms relevant to cognitive difficulties (n=7) and decreased FA values at the initial imaging time point for the 26 patients (Groups A and C)